

Paper Review:

Simultaneous clustering and estimation of heterogeneous graphical models

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August 19, 2025

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· Hao, B., Sun, W. W., Liu, Y., & Cheng, G. (2018).
Simultaneous clustering and estimation of heterogeneous graphical models.
Journal of Machine Learning Research, 18(217), 1-58.

Main contributions and outline

Main contributions

- Propose a general framework of **Simultaneous Clustering And estimation** of heterogeneous graphical models (SCAN).
- Develop a **non-asymptotic statistical analysis** for the output directly from the SCAN algorithm.
- Verify the proposed algorithm through **numerical experiments and real data analysis**.

- Methodology
- Population-based analysis
- Sample-based analysis
- Numerical experiments
- Real data
- Conclusion

Methodology

Heterogeneous graphical models

- Among K groups, samples in the k -th group are generated i.i.d. from the following **Gaussian distribution**:

$$f_k(x; \mu_k, \Sigma_k) = (2\pi)^{-p/2} |\Sigma_k|^{-1/2} \exp \left\{ -\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right\}.$$

- Let $\Omega_k = \Sigma_k^{-1}$ be the k -th precision matrix with the ij -th entry ω_{kij} .
- We write $\Theta_k = \text{vec}(\mu_k, \Omega_k) \in \mathbb{F}_k \subset \mathbb{R}^{p+p^2}$ as its vectorized representation, and write the parameter of interest Θ as

$$\Theta = (\Theta_1, \dots, \Theta_K)^T \in \mathbb{F} = \mathbb{F}_1 \times \dots \times \mathbb{F}_K \subset \mathbb{R}^{K(p+p^2)}.$$

- Note that the degrees of freedom of Θ are $K(p + p(p + 1)/2)$.
- Thus, X follows a K -heterogeneous graphical model when its density is defined as

$$X \sim f(x; \pi, \Theta) = \sum_{k=1}^K \pi_k f_k(x; \Theta_k), \quad \pi = (\pi_1, \dots, \pi_K), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

Objective function

- Consider the **penalized** log-likelihood function for the observed data

$$\log \mathcal{L}(\pi, \Theta; X_{1:n}) = \frac{1}{n} \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k f_k(X_i; \mu_k, \Omega_k^{-1}) \right) - \mathcal{R}(\Theta),$$

where $\mathcal{R}(\Theta)$ is a penalized term.

- This paper takes

$$\mathcal{R}(\Theta) = \underbrace{\lambda_1 \sum_{k=1}^K \|\mu_k\|_1}_{\mathcal{P}_1(\mu)} + \underbrace{\lambda_2 \sum_{k=1}^K \|\Omega_k\|_{1,\text{off}}}_{\mathcal{P}_2(\Omega)} + \underbrace{\lambda_3 \sum_{i \neq j} \left(\sum_{k=1}^K \omega_{kij}^2 \right)^{1/2}}_{\mathcal{P}_3(\Omega)}.$$

- The SCAN method aims to solve $\max_{\pi_k, \mu_k, \Omega_k} \log \mathcal{L}(\pi, \Theta; X_{1:n})$.

E-step

- By introducing the latent random vector $L = (L_1, \dots, L_K) \sim \text{Multi}(1, (\pi_1, \dots, \pi_K)^T)$, the **complete penalized log-likelihood** can be written as

$$\log \mathcal{L}^c(\pi, \Theta; X_{1:n}, L_{1:n}) = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K L_{ik} \left[\log \pi_k + \log f_k(X_i; \Theta_k) \right] - \mathcal{R}(\Theta).$$

- For the t -th E-step, taking the **conditional expectation with respect to the observed data** $X_{1:n}$ yields

$$E_{\pi^{(t-1)}, \Theta^{(t-1)}} \left(\log \mathcal{L}^c(\pi, \Theta; X_{1:n}, L_{1:n}) \mid X_{1:n} \right) = Q_n(\pi, \Theta \mid \pi^{(t-1)}, \Theta^{(t-1)}) - \mathcal{R}(\Theta),$$

where

$$Q_n(\pi, \Theta \mid \pi^{(t-1)}, \Theta^{(t-1)}) = \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K L_{ik}^{(t-1)} \log \pi_k}_{Q_n(\pi \mid \pi^{(t-1)}, \Theta^{(t-1)})} + \underbrace{\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K L_{ik}^{(t-1)} \log f_k(X_i; \Theta_k)}_{Q_n(\Theta \mid \pi^{(t-1)}, \Theta^{(t-1)})}, \quad \text{and}$$

$$L_{ik}^{(t-1)} = E_{\pi^{(t-1)}, \Theta^{(t-1)}}(L_{ik} \mid X_{1:n}) = \frac{\pi_k^{(t-1)} f_k(X_i; \Theta_k^{(t-1)})}{\sum_{\ell=1}^K \pi_{\ell}^{(t-1)} f_{\ell}(X_i; \Theta_{\ell}^{(t-1)})}.$$

- A typical t -th M-step is to solve

$$(\pi^{(t)}, \Theta^{(t)}) = \arg \max_{\pi, \Theta} Q_n(\pi, \Theta \mid \pi^{(t-1)}, \Theta^{(t-1)}) - \mathcal{R}(\Theta),$$

which **does not give a closed-form solution**.

- The t -th CM-step alternatively obtains a solution via **block-wise update**, that is,

$$\pi^{(t)} = \arg \max_{\pi} Q_n(\pi \mid \pi^{(t-1)}, \Theta^{(t-1)}),$$

$$\mu^{(t)} = \arg \max_{\mu} Q_n(\mu, \Omega^{(t-1)} \mid \pi^{(t-1)}, \Theta^{(t-1)}) - \lambda_1 \mathcal{P}_1(\mu),$$

$$\Omega^{(t)} = \arg \max_{\Omega} Q_n(\mu^{(t-1)}, \Omega \mid \pi^{(t-1)}, \Theta^{(t-1)}) - \lambda_2 \mathcal{P}_2(\Omega) - \lambda_3 \mathcal{P}_3(\Omega). \quad (1)$$

- The solution for (1) can be found efficiently by the ADMM algorithm, which **does not guarantee the symmetry** of the solutions in general.
- Thus, it requires an additional symmetrization strategy: $\omega_{kij}^{(t)} = \min(\omega_{kij}^{(t)}, \omega_{kji}^{(t)})$.

Population-based analysis

Notations

- Let $\mathcal{D}$ be the index set of diagonal components of K precision matrices, i.e., $\Theta_{\mathcal{D}} = (\omega_{111}, \dots, \omega_{1pp}, \dots, \omega_{K11}, \dots, \omega_{Kpp})$.
- Denote $\mathcal{O}$ as the complete index set of Θ and $\mathcal{D}^c = \mathcal{O} \setminus \mathcal{D}$ as the complement set of $\mathcal{D}$.
- Let $S_{\mu^*} = \bigcup_{k=1}^K \{j : \mu_{kj}^* \neq 0, j = 1, \dots, p\}$ and $S_{\Omega^*} = \bigcup_{k=1}^K \{(i, j) : \omega_{kij}^* \neq 0, i \neq j\}$, where μ^* and Ω^* are the true parameters.
- Define the sparsity parameters as $s_{\Omega^*} = |S_{\Omega^*}|$ and $s_{\mu^*} = |S_{\mu^*}|$.
- Let $\Xi \subset \mathbb{F}$ be some non-empty **convex** set of parameters, of which the interior contains Θ^* , and define the support space $\mathcal{M}$ as

$$\mathcal{M} = \{V \in \Xi : \mu_{kj} = 0 \text{ for all } j \notin S_{\mu^*}, \text{ and } \omega_{kij} = 0 \text{ for all } (i, j) \notin S_{\Omega^*}\}.$$

Population-based analysis

- A population counterpart for $Q_n(\pi, \Theta \mid \pi^{(t-1)}, \Omega^{(t-1)})$ can be defined as

$$Q(\pi', \Theta' \mid \pi, \Theta) = \underbrace{E_{\pi^*, \Theta^*} \left[\sum_{k=1}^K L_{\pi, \Theta, k}(X) \log \pi'_k \right]}_{Q(\pi' \mid \pi, \Theta)} + \underbrace{E_{\pi^*, \Theta^*} \left[\sum_{k=1}^K L_{\pi, \Theta, k}(X) \log f_k(X; \Theta') \right]}_{Q(\Theta' \mid \pi, \Theta)},$$

where $L_{\pi, \Theta, k}(X) = E_{\pi, \Theta}(L_k \mid X) \propto \pi_k f_k(X; \Theta_k)$.

- In the sequel, π is fixed at $\pi_1 = \dots = \pi_K = 1/K$, so π is omitted.
- (Lemma 5)** One can readily obtain $\Theta^M = \arg \max_{\Theta'} Q(\Theta' \mid \Theta)$ as

$$\mu_k^M = \frac{E_{\Theta^*}(L_{\Theta, k}(X)X)}{E_{\Theta^*}(L_{\Theta, k}(X))}, \quad \Omega_k^M = E_{\Theta^*}(L_{\Theta, k}(X)) \left[E_{\Theta^*}(L_{\Theta, k}(X)(X - \mu_k^M)(X - \mu_k^M)^T) \right]^{-1}.$$

Population-based analysis

Condition 6 (Sufficiently separable condition).

We assume K clusters are **sufficiently separable** such that given an appropriate small parameter $\gamma > 0$, it holds a.s.

$$L_{\Theta,k}(X) \cdot L_{\Theta,j}(X) \leq \frac{\gamma}{24(K-1) \sqrt{\max\{W, W', W''\}}}$$

for each pair $j \neq k$ and any $\Theta \in \mathcal{B}_\alpha(\Theta^*; \Xi) = \mathcal{B}_\alpha(\Theta^*) \cap \Xi$, where $\mathcal{B}_\alpha(\Theta^*) \subset \mathbb{F}$ is an α -radius closed ball centered at Θ^* w.r.t. $\|\cdot\|_2$.

- Recall the fact that $L_{\Theta,k}(X) = E_\Theta(L_k | X) \propto f_k(X; \Theta_k)$.
- Intuitively, Condition 6 ensures that $f_k(X; \Theta_k) \cdot f_j(X; \Theta_j) \approx 0$ whenever $\Theta \approx \Theta^*$.

Lemma 7 (Gradient stability).

Under Condition 6, the function $Q(\cdot \mid \Theta)$ satisfies

$$\|\nabla Q(\Theta^* \mid \Theta) - \nabla Q(\Theta^* \mid \Theta^*)\|_2 \leq \tau \cdot \|\Theta - \Theta^*\|_2,$$

with parameter $\tau \leq \gamma/12$ for any $\Theta \in \mathcal{B}_\alpha(\Theta^*; \Xi)$. The gradient $\nabla Q(\Theta^* \mid \Theta)$ is taken with respect to the first variable of $Q(\cdot \mid \cdot)$.

- Lemma 7 guarantees that the curvature of $Q(\cdot \mid \Theta)$ is similar to that of $Q(\cdot \mid \Theta^*)$ when Θ is close to Θ^* .

Sample-based analysis

Restricted strong concavity

Condition 8 (Bounded eigenvalue condition).

There exist some positive constants β_1, β_2 such that

$$0 < \beta_1 < \min_{1 \leq k \leq K} \lambda_{\min}(\Omega_k) < \max_{1 \leq k \leq K} \lambda_{\max}(\Omega_k) < \beta_2.$$

Lemma 9 (Restricted strong concavity).

Suppose that Condition 8 holds. Then, for any $\Theta \in \mathcal{B}_\alpha(\Theta^*)$, with probability at least $1 - \delta$, each $\Theta' \in \mathcal{B}_{2\alpha}(\Theta^*)$ satisfies

$$Q_n(\Theta' | \Theta) - Q_n(\Theta^* | \Theta) - \langle \nabla Q_n(\Theta^* | \Theta), \Theta' - \Theta^* \rangle \leq -\frac{\gamma}{2} \|\Theta' - \Theta^*\|_2^2,$$

with sufficiently large n , where $\gamma = c \cdot \min\{\beta_1, 0.5(\beta_2 + 2\alpha)^{-2}\}$ is the strong concavity parameter for some constant c .

Bound for gradient gap

Condition 10 (Bound for gradient gap).

For any fixed $\Theta \in \mathcal{B}_\alpha(\Theta^*; \Xi)$, with probability at least $1 - \delta_1$,

$$\left\| \left[\nabla Q_n(\Theta^* | \Theta) - \nabla Q(\Theta^* | \Theta) \right]_{\mathcal{D}^c} \right\|_{\mathcal{P}^*} \leq \epsilon_1,$$

and with probability at least $1 - \delta_2$, we have

$$\left\| \left[\nabla Q_n(\Theta^* | \Theta) - \nabla Q(\Theta^* | \Theta) \right]_{\mathcal{D}} \right\|_2 \leq \epsilon_2,$$

where ϵ_1 and ϵ_2 are functions of n , p , K , δ_1 , and δ_2 , and $\|\cdot\|_{\mathcal{P}^*}$ is the dual norm of $\mathcal{P}(\Theta) = M_1 \mathcal{P}_1(\mu) + M_2 \mathcal{P}_2(\Omega) + M_3 \mathcal{P}_3(\Omega)$ for some positive constants M_1 , M_2 , and M_3 .

Statistical error versus optimization error

Definition 11 (Support space compatibility constant).

For the support subspace $\mathcal{M} \subset \mathbb{R}^{K(p+p^2)}$, we define

$$\nu(\mathcal{M}) = \sup_{\Theta \in \mathcal{M} \setminus \{0\}} \frac{\mathcal{P}(\Theta)}{\|\Theta\|_2} = \sup_{\Theta \in \mathcal{M}, \|\Theta\|_2=1} \mathcal{P}(\Theta).$$

- Intuitively, $\nu(\mathcal{M})$ measures the **aggression** of $\mathcal{P}(\cdot)$ on $\mathcal{M}$.
- As the true parameters become denser, $\nu(\mathcal{M})$ tends to increase.
- Thus, the larger $\nu(\mathcal{M})$ is, the more samples are needed to achieve statistical consistency.

Statistical error versus optimization error

Theorem 13 (Convergence rate).

Suppose Conditions 6, 8, and 10 hold. Let $\kappa = 6\tau/\gamma \leq 1/2$, where τ, γ are calculated in Lemma 7 and Lemma 9. Consider the SCAN algorithm with initialization $\Theta^{(0)}$ falling into a ball $\mathcal{B}_\alpha(\Theta^*; \Xi)$ for some constant radius $\alpha > 0$ and assume the tuning parameters satisfy $\lambda_1 = M_1 \lambda_n^{(t)}$, $\lambda_2 = M_2 \lambda_n^{(t)}$, $\lambda_3 = M_3 \lambda_n^{(t)}$, and

$$\lambda_n^{(t)} = \epsilon + \frac{\kappa}{\gamma} \nu(\mathcal{M}) \|\Theta^{(t-1)} - \Theta^*\|_2.$$

If the sample size n is large enough such that $\epsilon \leq (1 - \kappa)\gamma\alpha/(6\nu(\mathcal{M}))$, then $\Theta^{(t)}$ satisfies, with probability at least $1 - t\delta'$,

$$\|\Theta^{(t)} - \Theta^*\|_2 \leq \underbrace{\frac{6\nu(\mathcal{M})}{(1 - \kappa)\gamma} \epsilon}_{\text{Statistical Error (SE)}} + \underbrace{\kappa^t \|\Theta^{(0)} - \Theta^*\|_2}_{\text{Optimization Error (OE)}},$$

where $\delta' = \delta + \delta_1 + \delta_2$ with $\delta, \delta_1, \delta_2$ defined in Lemma 9 and Condition 10 and $\epsilon = \epsilon_1 + \epsilon_2/\nu(\mathcal{M})$.

Consistency

Condition 16.

The largest element of cluster means and precision matrices are both bounded, that is, for some positive constants c_1 and c_2 ,

$$\|\mu^*\|_\infty = \max_{1 \leq k \leq K} \|\mu_k^*\|_\infty < c_1, \quad \text{and} \quad \|\Omega^*\|_\infty = \max_{1 \leq k \leq K} \|\Omega_k^*\|_\infty < c_2.$$

Condition 17.

Suppose that the number of clusters K satisfies $K^2 = o(p(\log n)^{-1})$.

Corollary 18 (Consistency).

Suppose Conditions 6, 8, 10, 16, and 17 hold. If the sample size n and the iteration step size T is sufficiently large, the optimization error is dominated by the statistical error, and thus

$$\sum_{k=1}^K \left(\left\| \mu_k^{(T)} - \mu_k^* \right\|_2 + \left\| \Omega_k^{(T)} - \Omega_k^* \right\|_F \right) \lesssim \underbrace{\left(\|\Omega^*\|_\infty \sqrt{\frac{K^5 s_{\mu^*} \log p}{n}} \right)}_{\text{Cluster means error}} + \underbrace{\left(\|\Omega^*\|_\infty \sqrt{\frac{K^3 (K s_{\Omega^*} + p) \log p}{n}} \right)}_{\text{Precision matrices error}}.$$

Support consistency

Remark 20.

To ensure the support consistency, we threshold the precision matrix estimator $\Omega_k^{(T)}$ such that $\tilde{\omega}_{kij} = \omega_{kij}^{(T)} I(|\omega_{kij}^{(T)}| > r_n)$ for $r_n = (\sqrt{K^5 s_{\mu^*}} + \sqrt{K^3 (K s_{\Omega^*} + p)}) \sqrt{(\log p)/n}$.

Under the minimal signal assumption for the true precision matrices, the support consistency can be accomplished with probability tending to one.

Theorem S.2 (Support consistency).

Denote the final precision estimator as $\tilde{\Omega}_k$ and its support as $\tilde{\mathcal{V}}_k$. Under the conditions in Corollary 18 and the minimal signal condition, we have, with probability tending to one, $\tilde{\mathcal{V}}_k = \mathcal{V}_k$, where $\mathcal{V}_k$ is the support of Ω_k^* .

Numerical experiments

Universal settings

- A cluster membership Y_i is uniformly sampled from $\{1, 2, 3\}$.
- Given the cluster label, each sample X_i is generated from a Gaussian distribution $N(\mu(Y_i), \Sigma(Y_i))$.
- The cluster mean $\mu(Y_i)$ is sparse: its first 10 variables are non-zero (defined based on Y_i), and the last $p - 10$ variables are zeros. Precisely,

$$\mu(Y_i) = (\mu \mathbf{1}_5^T, -\mu \mathbf{1}_5^T, 0)^T I(Y_i = 1) + (\mu \mathbf{1}_{10}^T, 0)^T I(Y_i = 2) + (-\mu \mathbf{1}_5^T, -\mu \mathbf{1}_5^T, 0)^T I(Y_i = 3).$$

- The number of features $p = 100$ and sample size $n = 300$ are used in the simulations.
- The initial values $\pi^{(0)}$, $\mu^{(0)}$, and $\Omega^{(0)}$ are obtained by random initialization.

Regular network

- **Regular network.**

- Each precision matrix Ω_j ($j = 1, 2, 3$) is a $p \times p$ matrix generated as a 5-block tridiagonal matrix.
- A $p \times p$ matrix $A = (a_{ij})$ is defined as tridiagonal if $a_{ij} = 0$ whenever $|i - j| > 1$.
- To allow the similarity across clusters, the off-diagonal entries of $\Omega_1, \Omega_2, \Omega_3$ are set to $\eta, 0.99\eta$, and 1.01η , respectively.
- All diagonal entries are 1.

- Model 1: $\mu = 0.8$ and $\eta = 0.3$.
- Model 2: $\mu = 1$ and $\eta = 0.3$.
- Model 3: $\mu = 1$ and $\eta = 0.4$.

Regular network

Models	Methods	CE	CME	PME	TPR /FPR
Model 1 $\mu = 0.8$ $\eta = 0.3$	<i>K</i> -means	0.166 _{0.011}	2.256 _{0.108}	NA	NA /NA
	<i>K</i> -means + JGL	0.166 _{0.011}	2.256 _{0.108}	8.206 _{0.090}	0.985 _{0.001} /0.023 _{0.001}
	Zhou et al. (2009)	0.104 _{0.007}	1.190 _{0.052}	10.458 _{0.0509}	0.960 _{0.002} /0.107 _{0.001}
	SCAN	0.071_{0.007}	1.120_{0.063}	7.620_{0.072}	0.993_{0.001} /0.022_{0.001}
Model 2 $\mu = 1$ $\eta = 0.3$	<i>K</i> -means	0.210 _{0.009}	3.428 _{0.114}	NA	NA /NA
	<i>K</i> -means + JGL	0.210 _{0.009}	3.428 _{0.114}	12.099 _{0.317}	0.989 _{0.001} /0.039 _{0.003}
	Zhou et al. (2009)	0.125 _{0.012}	1.860 _{0.118}	12.833 _{0.253}	0.993 _{0.001} /0.119 _{0.006}
	SCAN	0.058_{0.012}	1.476_{0.145}	10.301_{0.332}	0.997_{0.001} /0.036_{0.002}
Model 3 $\mu = 1$ $\eta = 0.4$	<i>K</i> -means	0.021 _{0.002}	1.289 _{0.013}	NA	NA /NA
	<i>K</i> -means + JGL	0.021 _{0.002}	1.289 _{0.013}	7.639 _{0.061}	0.993 _{0.001} /0.029 _{0.002}
	Zhou et al. (2009)	0.021 _{0.002}	0.968 _{0.018}	10.115 _{0.047}	0.968 _{0.001} /0.106 _{0.001}
	SCAN	0.014_{0.001}	0.956_{0.018}	7.614_{0.061}	0.993_{0.001} /0.029_{0.002}

Power-law network

- **Power-law network.**

- Each of the networks has ten equally sized unconnected subnetworks.
- The entries of the initial $p \times p$ matrix w_{1ij} are defined as:
 - $w_{1ij} = 1$ if $i = j$ (diagonal entries).
 - $w_{1ij} = 0$ if $i \neq j$ and there is no edge (explicit sparsity).
 - $w_{1ij} \sim \text{Unif}([-0.4, -0.1] \cup [0.1, 0.4])$ if $i \neq j$ and an edge exists (non-zero values for connections).
- Eight of the ten subnetworks share identical structure and edge values across all three clusters.
- One subnetwork remains the same for only the first two classes, and the last one appears only for the first class.

- Model 4: $\mu = 0.7$.
- Model 5: $\mu = 0.8$.
- Model 6: $\mu = 0.9$.

Power-law network

Models	Methods	CE	CME	PME	TPR /FPR
Model 4 $\mu = 0.7$	<i>K</i> -means	0.331 _{0.007}	3.282 _{0.047}	NA	NA /NA
	<i>K</i> -means + JGL	0.331 _{0.007}	3.282 _{0.047}	49.516 _{0.159}	0.575 _{0.002} /0.034 _{0.002}
	Zhou et al. (2009)	0.311 _{0.006}	2.494 _{0.055}	50.945 _{0.164}	0.578 _{0.002} /0.134 _{0.002}
	SCAN	0.283_{0.008}	2.385_{0.065}	48.845_{0.146}	0.577 _{0.003} /0.032 _{0.002}
Model 5 $\mu = 0.8$	<i>K</i> -means	0.228 _{0.010}	2.777 _{0.111}	NA	NA/NA
	<i>K</i> -means + JGL	0.228 _{0.010}	2.777 _{0.111}	48.601 _{0.132}	0.582 _{0.002} /0.044 _{0.003}
	Zhou et al. (2009)	0.186 _{0.011}	1.837 _{0.113}	49.289 _{0.122}	0.584 _{0.001} /0.131 _{0.001}
	SCAN	0.156_{0.012}	1.789_{0.119}	47.729_{0.118}	0.583 _{0.002} /0.041 _{0.002}
Model 6 $\mu = 0.9$	<i>K</i> -means	0.083 _{0.010}	1.624 _{0.120}	NA	NA /NA
	<i>K</i> -means + JGL	0.083 _{0.010}	1.624 _{0.120}	46.879 _{0.093}	0.589 _{0.002} /0.070 _{0.003}
	Zhou et al. (2009)	0.050 _{0.002}	1.003 _{0.018}	47.503 _{0.003}	0.591 _{0.001} /0.128 _{0.001}
	SCAN	0.045_{0.002}	1.003_{0.018}	46.356_{0.086}	0.589 _{0.001} /0.068 _{0.003}

- **Chain network.**

- Similar to the power-law network, each of the networks consists of ten unconnected subnetworks.
- The ij -th element of each block of the first covariance matrix Σ_1 is constructed as $\sigma_{ij} = \exp(-|s_i - s_j|)$, $s_1 < s_2 < \dots < s_{p/10}$ with $s_i - s_{i-1} \sim \text{Unif}(0.5, 1)$.
- Σ_2 is created to be identical to Σ_1 but with one of the ten block matrices reset to an identity matrix.
- Similarly, Σ_3 resets an additional block matrix besides the one reset for Σ_2 .

- Model 7: $\mu = 0.7$.
- Model 8: $\mu = 0.8$.
- Model 9: $\mu = 0.9$.

Chain network

Models	Methods	CE	CME	PME	TPR /FPR
Model 7 $\mu = 0.7$	<i>K</i> -means	0.277 _{0.005}	2.705 _{0.070}	NA	NA /NA
	<i>K</i> -means + JGL	0.277 _{0.005}	2.705 _{0.070}	25.608 _{0.183}	0.995 _{0.000} /0.033 _{0.001}
	Zhou et al. (2009)	0.267 _{0.006}	1.815 _{0.075}	29.341 _{0.109}	0.991 _{0.001} /0.131 _{0.002}
	SCAN	0.231_{0.007}	1.652_{0.087}	25.110_{0.106}	0.991 _{0.001} /0.031 _{0.001}
Model 8 $\mu = 0.8$	<i>K</i> -means	0.200 _{0.008}	2.124 _{0.098}	NA	NA/NA
	<i>K</i> -means + JGL	0.200 _{0.008}	2.124 _{0.098}	24.499 _{0.127}	0.996 _{0.000} /0.042 _{0.001}
	Zhou et al. (2009)	0.168 _{0.004}	1.055 _{0.076}	27.494 _{0.121}	0.995 _{0.001} /0.131 _{0.001}
	SCAN	0.140_{0.004}	1.046_{0.038}	23.804_{0.085}	0.996 _{0.000} /0.039 _{0.001}
Model 9 $\mu = 0.9$	<i>K</i> -means	0.123 _{0.005}	1.465 _{0.040}	NA	NA /NA
	<i>K</i> -means + JGL	0.123 _{0.005}	1.465 _{0.040}	23.663 _{0.097}	0.997 _{0.000} /0.044 _{0.001}
	Zhou et al. (2009)	0.116 _{0.003}	1.031 _{0.022}	26.476 _{0.090}	0.996 _{0.001} /0.131 _{0.001}
	SCAN	0.098_{0.003}	1.025_{0.022}	23.425_{0.083}	0.998 _{0.000} /0.043 _{0.002}

Real data

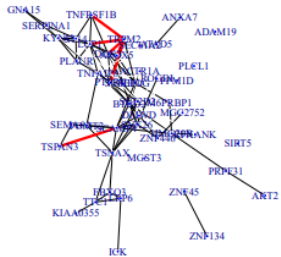
Glioblastoma cancer data

- The raw gene expression dataset measures 17,814 levels of mRNA expression of $n = 482$ GBM patients.
- Each patient belongs to one of $K = 4$ subgroups of GBM: Classical, Mesenchymal, Neural, and Proneural.
- Among the 17,814 variables, 840 signature genes are considered, from which $p = 50$ are finally selected after the sure independence screening analysis.
- The tuning parameters are set at $\lambda_1 = 0.065$, $\lambda_2 = 0.238$, and $\lambda_3 = 0.138$.

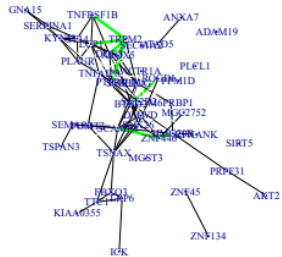
Results

Methods	Clustering Error	$\sum_k \ \hat{\boldsymbol{\mu}}^{(k)}\ _0$	$\sum_k \ \hat{\boldsymbol{\Omega}}^{(k)}\ _0$
<i>K</i> -means	0.262	200	NA
Zhou et al. (2009)	0.336	106	1820
<i>K</i> -means + JGL	0.262	200	1360
SCAN	0.222	128	1452

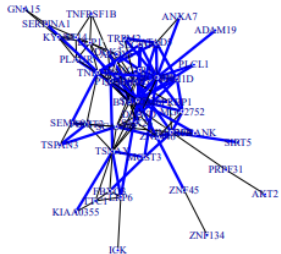
Results



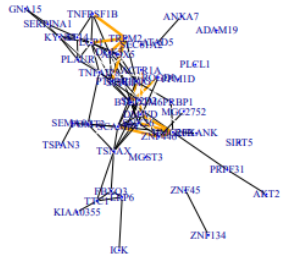
Classical



Mesenchymal



Neural



Proneural

Conclusion

Conclusion

- This paper proposes a new SCAN method for **detecting heterogeneous graphical models**.
- Its theoretical guarantees are explored, which addresses the **trade-off between statistical accuracy and computational complexity**.
- The assumption that **the initial value is close enough to the true parameter** resolves several technical issues.
- The **support consistency** remains vague.
- The empirical results betray that **the estimators tend to be sparse**:

$$\Omega^{(t)} = \arg \max_{\Omega} Q_n(\mu^{(t-1)}, \Omega \mid \pi^{(t-1)}, \Theta^{(t-1)}) - \lambda_2 \mathcal{P}_2(\Omega) - \lambda_3 \mathcal{P}_3(\Omega).$$

References

- Hao, B., Sun, W. W., Liu, Y., & Cheng, G. (2018). Simultaneous clustering and estimation of heterogeneous graphical models. *Journal of Machine Learning Research*, 18(217), 1-58.